

Phase Separation of the Nup62 FG-Repeat Domain: Towards a Mechanistic Basis for Nuclear Pore Complex Selectivity

Overview

The nuclear pore complex (NPC) is the sole gateway for macromolecular traffic between the nucleus and cytoplasm. Its selectivity is thought to arise from a barrier formed by the intrinsically disordered, phenylalanine-glycine (FG) repeat domains of nucleoporins that project into the central channel (Fig. 1). Among these, Nup62 is a core component of the central channel itself, contacts essentially every cargo that crosses the pore, and has been implicated in several diseases, including neurodegenerative and developmental disorders making it both a mechanistically central and clinically relevant target. My thesis tested the hypothesis that the FG-repeat intrinsically disordered region (IDR) of human Nup62 undergoes liquid–liquid phase separation (LLPS), and that this phase-separated state contributes to the selective permeability of the channel, rather than the FG domain acting merely as a sequence-generic, passive spacer, as disordered regions are conventionally assumed to be.

Starting from a computational identification of the Nup62 FG-IDR and a cross-species conservation analysis of its FG motifs, I moved to recombinant expression and purification of the domain a technically demanding target with a near-zero extinction coefficient and used a serendipitous purification finding — that the domain failed to elute cleanly from size-exclusion chromatography in standard-salt buffer, apparently due to on-column LLPS, and required a no-salt buffer to elute as a single species — to motivate a novel, centrifugation-based phase-separation assay. To reach the protein concentrations needed for physiologically meaningful condensate assays, I developed a reversed-phase HPLC and lyophilization protocol to concentrate the domain into the millimolar concentration, ran an initial exploratory in vitro condensate-formation experiment, and validated the imaging pipeline itself against a well-established BSA/PEG positive control. In parallel, I purified NTF2 (a physiological NPC transport receptor) and mEGFP (an inert control) as the reagents needed for a future selective-partitioning assay, to directly test whether Nup62 condensates reproduce the NPC's hallmark cargo selectivity.

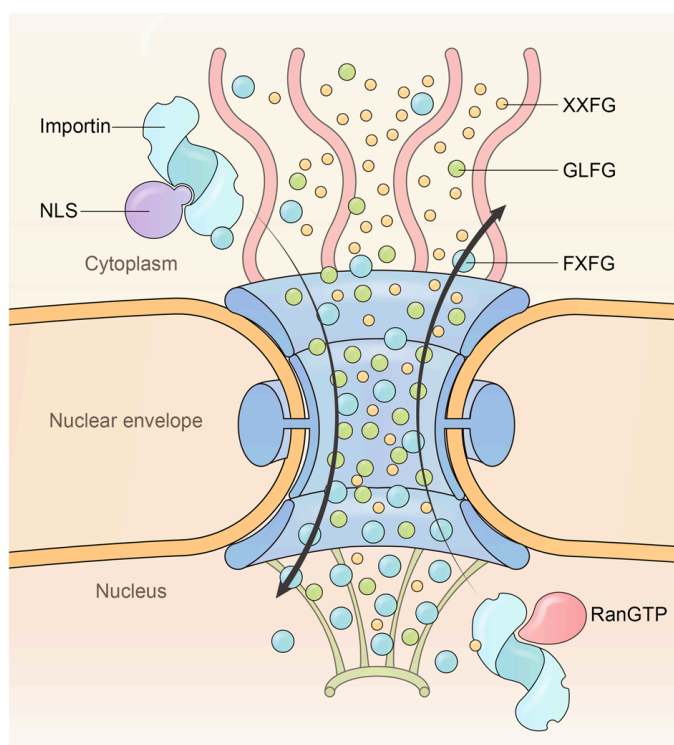


Figure 1. The nuclear pore complex and its FG-Nup permeability barrier. Transport receptors (e.g., importins) shuttle cargo bearing nuclear localization signals (NLS) through the FG-repeat meshwork of the central channel, a process regulated by the RanGTP gradient.

Background and Aims

FG-Nups form the selective permeability barrier of the NPC: small molecules diffuse through relatively freely, while larger cargo requires active, transport-receptor-mediated passage. Several models describe how this barrier works mechanistically, including a selective-phase model in which FG domains form a cohesive, condensate-like meshwork. Peripheral FG-Nups such as Nup98 have already been shown to phase-separate in vitro. Nup62, by contrast, sits within the central transport channel itself and is recurrently implicated in neurodegenerative and developmental disease, making it an especially interesting candidate for testing whether phase separation of a central-channel FG domain, specifically, contributes to selectivity.

Because IDRs are conventionally treated as sequence-generic spacers evolving under weak constraint, I first asked whether the Nup62 FG-IDR shows any detectable sequence-level constraint at all, before testing its biophysical behavior directly.

Specific aims:

- Computationally identify the intrinsically disordered, FG-repeat-containing region of human Nup62.
- Determine whether this FG-IDR shows evolutionary conservation across vertebrates, and characterize any sequence-level patterning within the repeats.
- Express and purify the human Nup62 FG-IDR, and develop a quantitative, fluorescence-independent assay for its phase-separation behaviour, necessitated by the domain's near-zero extinction coefficient.

- Establish a route to concentrate the purified domain to the millimolar range required for in vitro condensate assays.
- Purify a physiological cargo (NTF2) and an inert probe (mEGFP) as reagents for a future selective-partitioning assay, to test whether Nup62 condensates retain NPC-like selectivity.

1. Computational Identification and Evolutionary Conservation of the Nup62 FG-IDR

I first computationally identified the intrinsically disordered, FG-repeat-containing region of human Nup62 (Fig. 2, bottom right, showing FG-repeat positions relative to the protein's C-terminal α -helical domain). Because disordered regions are generally assumed to evolve neutrally, tolerating substitutions as long as gross physicochemical properties such as charge and hydrophobicity are preserved, I next asked whether this holds for the Nup62 FG-IDR specifically, as the IDR region is the one physically spanning the central transport channel. Interacting Physically with cargo

A multiple sequence alignment of the FG-IDR across vertebrate species, scored per residue for conservation, did not fit the 'neutral spacer' expectation. Conservation was not uniformly low across the disordered region; instead, conserved residues (score 6–9) were concentrated at specific positions within each FG-repeat motif, predominantly the phenylalanine and glycine themselves, while the intervening spacer sequence remained comparatively variable (Fig. 2). This was a first indication that, despite being classified as intrinsically disordered, this region is not evolving as a generic spacer.

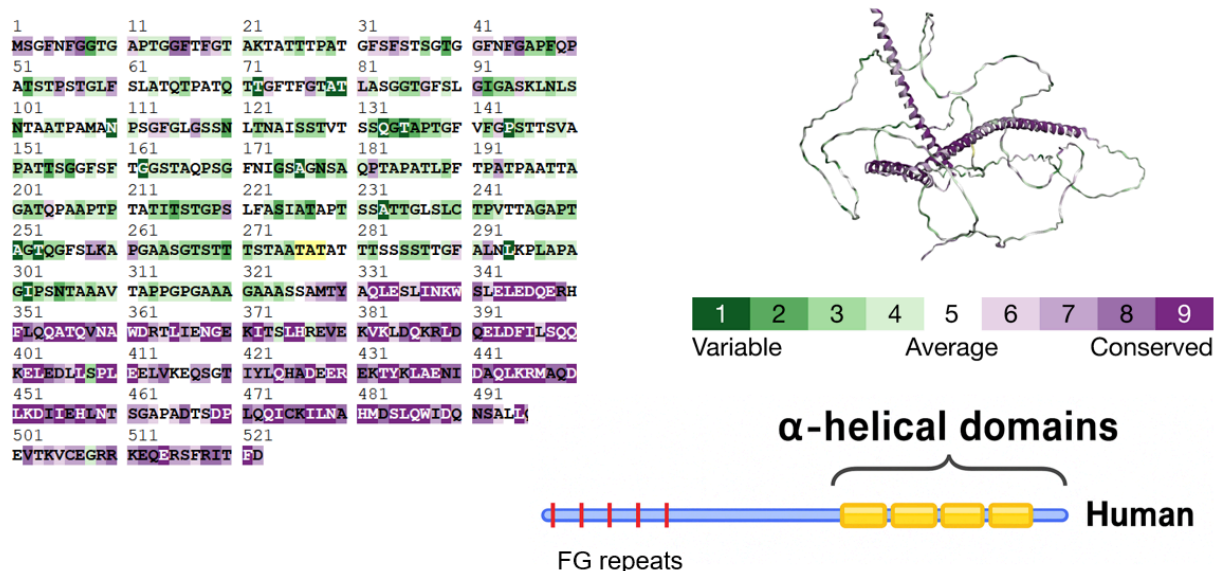


Figure 2. Per-residue conservation across the aligned Nup62 FG-IDR (left), mapped onto a predicted structural model (top right; 1 = variable, 9 = conserved), and onto the domain's linear architecture, showing FG-repeat positions relative to the C-terminal α -helical region (bottom right).

To interrogate this further, I wrote code to analyze amino acid adjacency across the aligned sequences, and found that phenylalanine and glycine reciprocally flank one another far more often than any other residue pair: glycine is by far the most frequent residue immediately preceding and immediately following each phenylalanine in the domain (Fig. 3). This is consistent with a canonical, sequence-constrained FG/GxFG-type repeat, rather than a loosely defined hydrophobic patch. Because the F and G positions are also the residues implicated in cohesive inter-repeat interactions in other FG-Nups,

this raised a specific, testable hypothesis: that this conservation reflects selection for the intermolecular contacts that drive condensation or barrier assembly, a redundant but functionally important role for phase behavior in Nup62, consistent with its central position in the channel rather than conservation for its own sake.

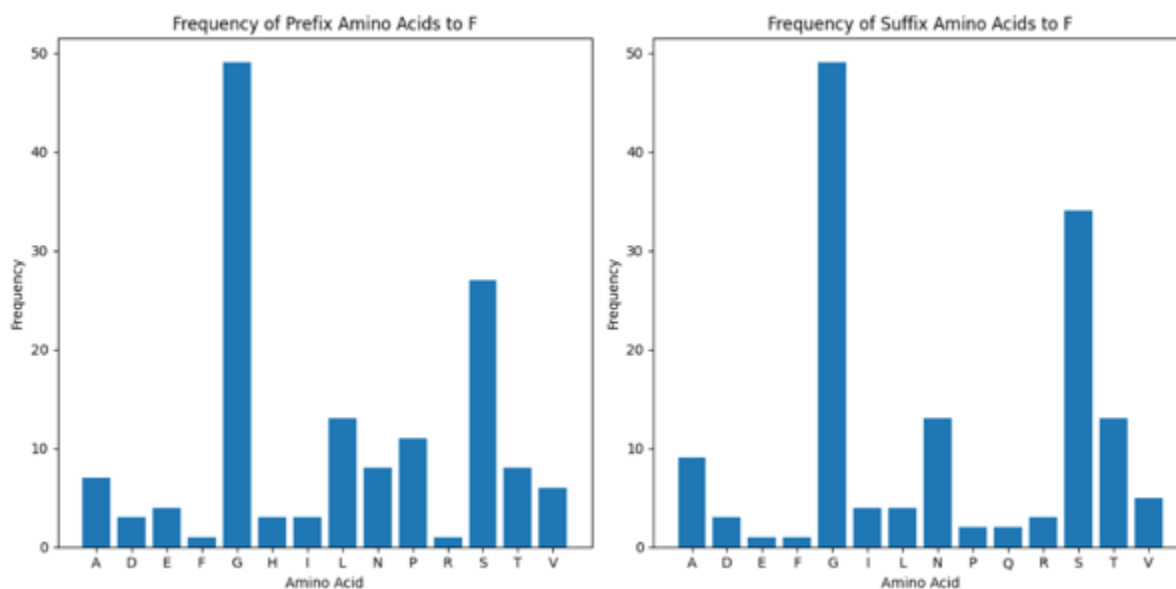


Figure 3. Frequency of the amino acid immediately preceding (left) and following (right) each phenylalanine (F) in the Nup62 FG-IDR. Glycine dominates both flanking positions.

2. Purification of the Nup62 FG-IDR: A Serendipitous Clue

Recombinant expression and purification of the human Nup62 FG-IDR proved unusually difficult. Because the domain is almost entirely devoid of aromatic residues outside the FG motifs themselves, its extinction coefficient is close to zero it is essentially invisible to standard UV-based detection and quantification. Every purification step, therefore had to be tracked indirectly, by collecting all eluted fractions and resolving them by SDS-PAGE, rather than by monitoring absorbance directly.

A more informative complication emerged during size-exclusion chromatography (SEC). In a standard SEC buffer (150–200 mM NaCl), the domain would not elute cleanly from the column — consistent with the protein undergoing phase separation on-column and interacting with the Superdex resin rather than passing through as a soluble species. Switching to a no-salt elution buffer resolved this: the domain eluted cleanly at the expected size. This indicated that even moderate, near-physiological ionic strength was sufficient to drive the domain's self-association, consistent with a classical salting-out mechanism given the domain's hydrophobic character. A literature search for precedent turned up reports that the peripheral FG-Nup Nup98 undergoes salt-induced phase separation, raising the possibility that the same mechanism applies to Nup62's central-channel FG-IDR, and directly motivating the assay described next.

3. A Quantitative, Salt-Induced Phase-Separation Assay

To test this hypothesis directly, I built a centrifugation-based phase-separation assay (Fig. 4): because phase-separated protein is denser than the surrounding buffer, it pellets under centrifugation and can be separated from soluble protein in the supernatant, then resolved and quantified by SDS-PAGE.

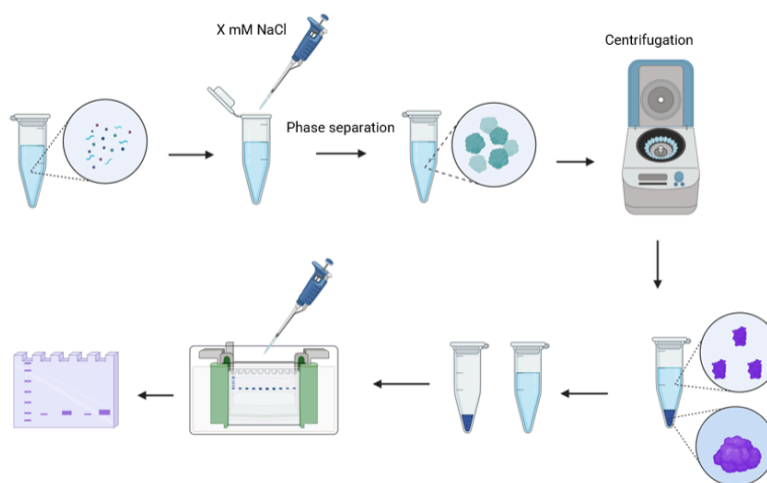


Figure 4. Workflow for the centrifugation-based phase-separation assay: induce phase separation, pellet by centrifugation, resolve supernatant and pellet fractions by SDS-PAGE.

Applying this assay across a range of NaCl concentrations showed a clear, salt-dependent shift of the Nup62 FG-IDR from the supernatant into the pellet fraction (Fig. 5), quantified by densitometry (ImageJ). This confirmed, quantitatively, the salt-induced condensation behavior first hinted at by the SEC anomaly above, and is consistent with a salting-out-driven condensation mechanism, in line with the hydrophobic character of the conserved FG motifs identified in Section 1

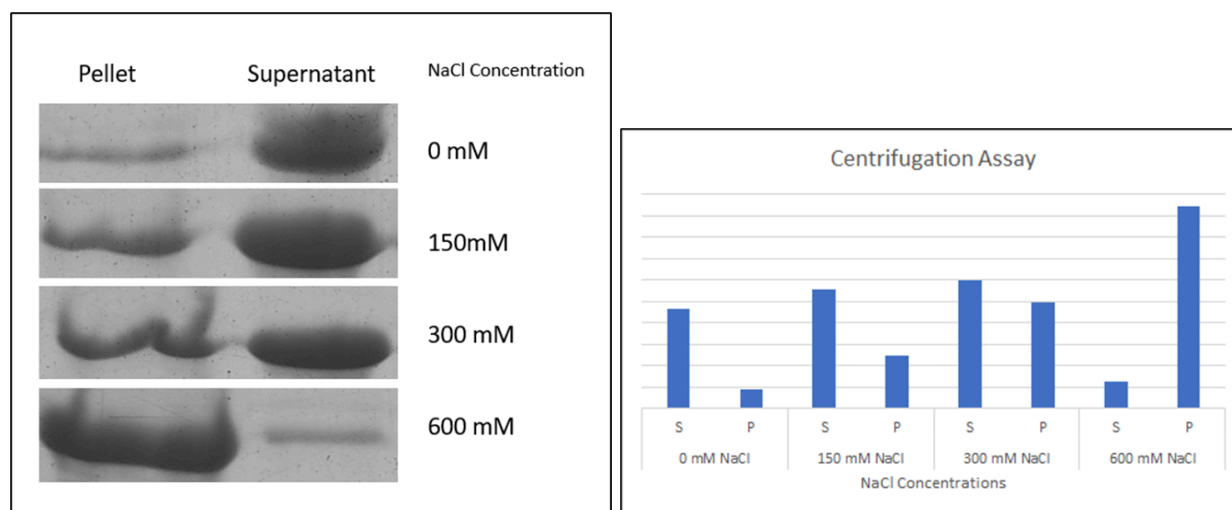


Figure 5. SDS-PAGE of pellet and supernatant fractions across a NaCl concentration series, and below, densitometric quantification of the supernatant (S) vs. pellet (P) signal at each condition.

4. From Hydrogels to Condensates: Designing a Selectivity Test

With salt-induced phase separation established, I next had to decide how to test whether Nup62 phase separation reproduces NPC-like selectivity. Görlich and colleagues have previously used cross-linked Nup98 hydrogels to demonstrate the selective passage of a transport receptor over an inert protein an elegant proof of principle, but hydrogels are covalently cross-linked, static structures that do not recapitulate the dynamic, liquid nature of a physiological FG barrier. I therefore chose to pursue biomolecular condensates rather than hydrogels, with the aim of directly challenging Nup62 condensates

with NTF2 (a physiological NPC transport receptor) against an inert control mEGFP, to test selective partitioning across the condensate.

5. Concentrating the Nup62 FG-IDR for In Vitro Condensate Assays

Testing this, however, first required forming Nup62 condensates at concentrations approaching those found physiologically at the NPC, typically in the millimolar range. This posed a second, distinct purification challenge because even moderate ionic strength was sufficient to trigger phase separation, standard concentration methods such as centrifugal filtration which locally concentrate protein at the filter membrane risked inducing the same on-membrane phase separation and clogging observed during SEC, rather than yielding a usable concentrated stock.

I addressed this by switching to reversed-phase HPLC, purifying the domain in acetonitrile, a volatile organic solvent and then lyophilizing the eluted protein using a SpeedVac to a dry powder. This powder could then be resuspended directly to the required millimolar concentrations, thereby circumventing the concentration bottleneck entirely. Establishing this route, reversed-phase purification, solvent exchange, and lyophilization of an aggregation-prone, hydrophobic IDR was slow and technically demanding. Given the one-year timeline of the thesis, it allowed for a single, exploratory in vitro condensate-formation experiment using the concentrated material, described next.

6. Microscopy: Aggregation vs. Liquid Condensation

To visualize the reconstituted Nup62 FG-IDR directly, I used a brightfield/DIC microscopy protocol. Under the conditions tested, the domain formed small, irregular particulates (Fig. 6) rather than the smooth, spherical, fusion-competent droplets that define a liquid condensate. Because this was both a first-time in vitro LLPS experiment for me and a single run given time constraints, I could not yet distinguish with confidence whether this reflected genuine aggregate-like behavior of Nup62 or a limitation of the imaging approach itself. Resolving that ambiguity required independently validating the pipeline, using a system in which the expected outcome was already well established.

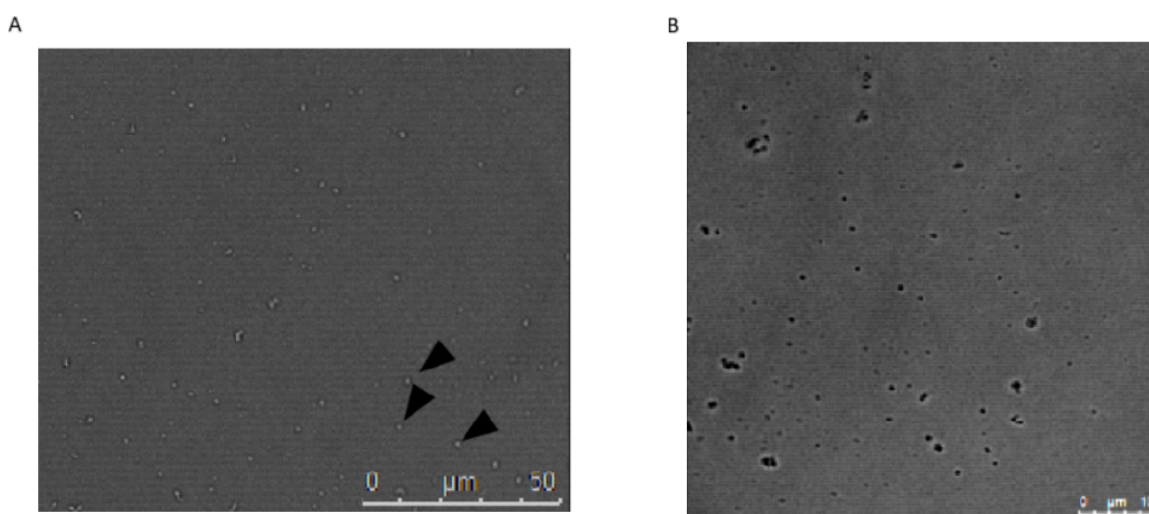


Figure 6. Brightfield microscopy of the Nup62 FG-IDR after inducing phase separation. Arrowheads (A) mark small particulate structures; panel B shows a higher-density field of similar particulates.

7. Validating the Imaging Pipeline: The BSA/PEG Positive Control

To master the *in vitro* condensate-formation and imaging workflow, and to confirm that the pipeline itself was working correctly, I reproduced a well-established positive control from the literature: BSA forming condensates under molecular crowding with 12% PEG (Fig. 7), visualized by DIC and, independently, by fluorescence using the non-covalent dye SYPRO Orange (Fig. 8) — including a no-protein negative control to confirm that the fluorescence signal was condensate-specific.

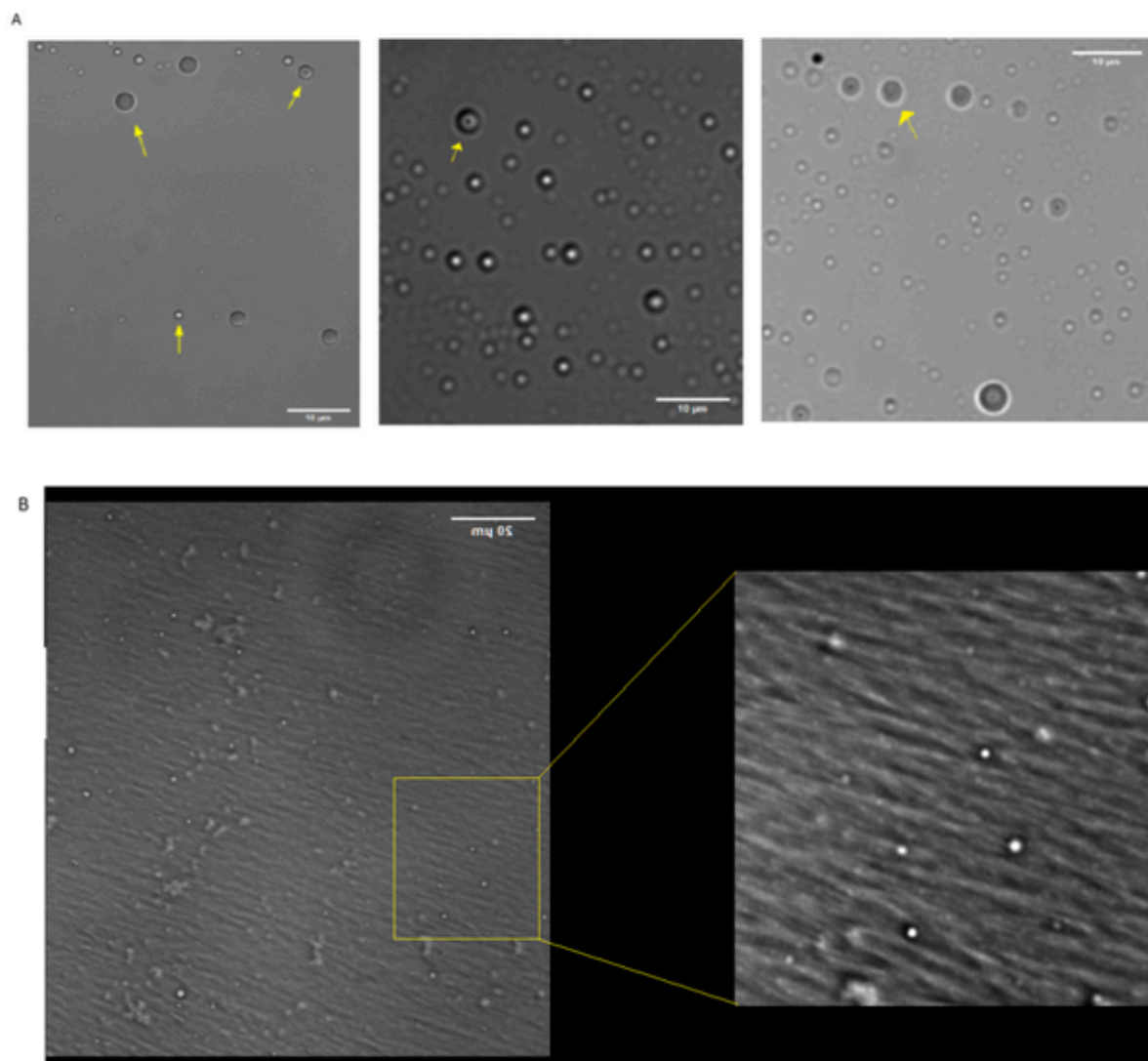


Figure 7. DIC micrographs of BSA (40 μM) condensates formed in 12% PEG at two magnifications (B, inset zoom).

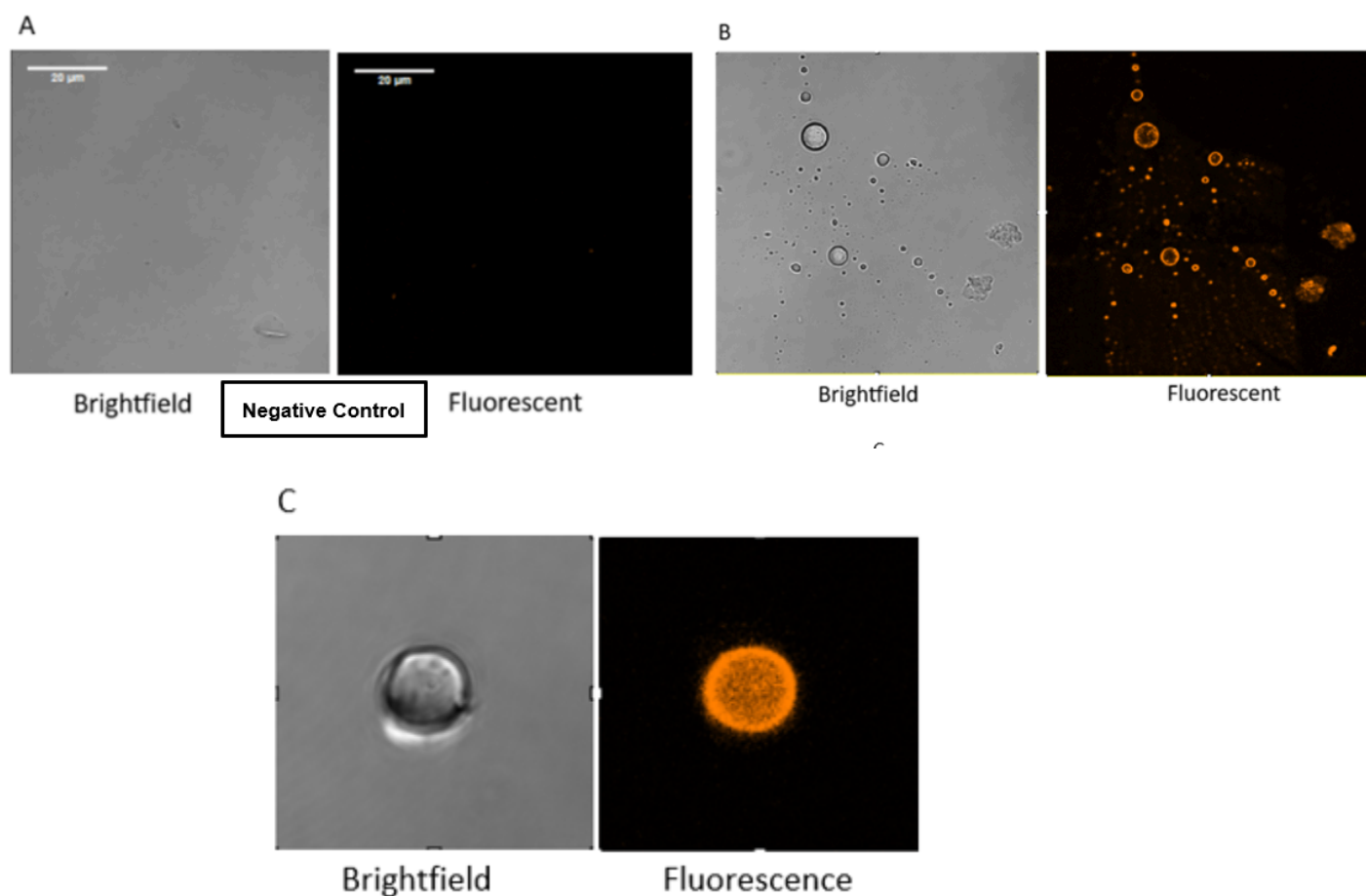


Figure 8. BSA/PEG condensates visualized by non-covalent SYPRO Orange labeling. (A) No-protein negative control shows no fluorescence signal; (B) BSA condensates show clear fluorophore partitioning into droplets; (C) zoomed brightfield and fluorescence imaging of a single condensate/particle.

Together, these controls show that the imaging pipeline reliably detects genuine liquid condensates when they are present. This strengthens, rather than undermines, the Nup62 result above: it indicates that the particulate, aggregate-like morphology observed for the Nup62 FG-IDR reflects the protein's actual behavior under the conditions tested, rather than a limitation of the method. Nevertheless, the *in vitro* imaging pipeline for Nup62 needs further optimization, as there were already positive results for salt-dependent phase separation from the Centrifugation assay.

8. Purification of NTF2 and mEGFP for a Future Selective-Partitioning Assay

The original aim of this step of the project was to test whether Nup62 condensates retain the NPC's hallmark selectivity *in vitro*, permitting entry of a physiological transport receptor while excluding an inert protein of similar size (Section 4). As a first, necessary step towards that experiment, I purified NTF2 (an NPC transport receptor that is actively and selectively transported across the FG barrier *in vivo*) and mEGFP (an inert fluorescent probe) as the two reagents required for this comparison (Fig. 11, Fig. 12).

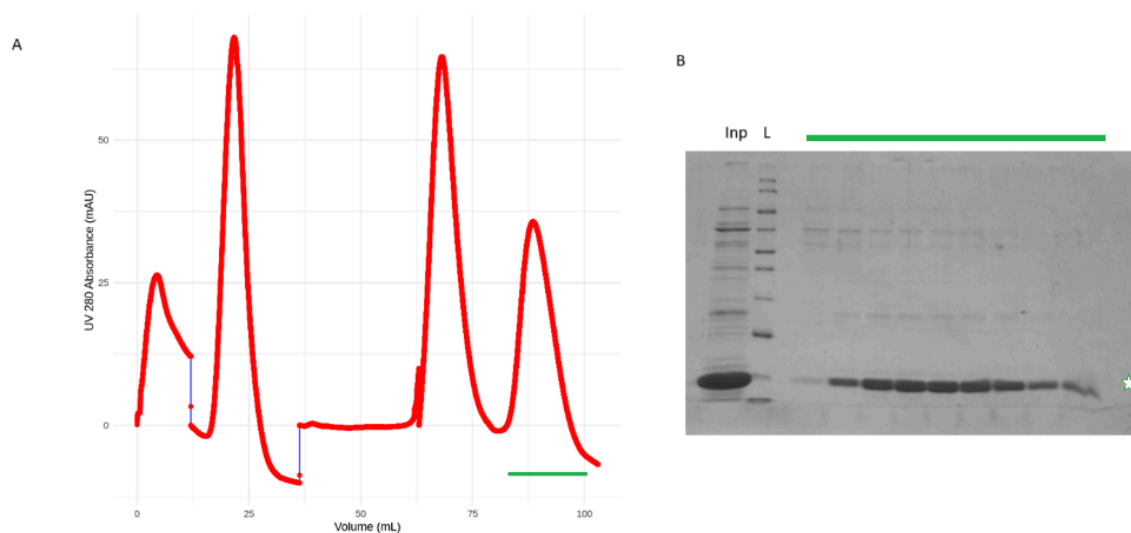


Figure 11. (A) Size-exclusion chromatogram of NTF2 (Superdex 75 16/600). (B) SDS-PAGE of the fractions under the collected peak, confirming a single, pure species.

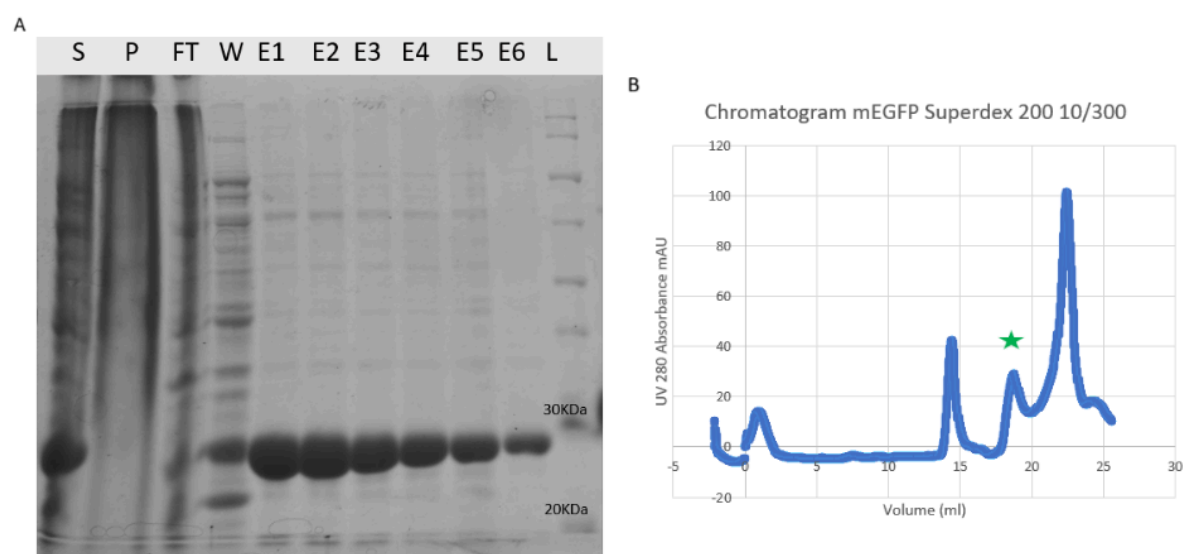


Figure 12. (A) SDS-PAGE profile of mEGFP (27.3 kDa) purification by Ni-NTA affinity chromatography (S: supernatant, P: pellet, FT: flow-through, W: wash, E1–E6: elutions). (B) Size-exclusion chromatogram (Superdex 200 10/300) of the pooled elutions; the monomeric peak is marked (★).

Both proteins were purified to a single, clean, monomeric species suitable for downstream use. As my contract ended before the Nup62 concentration pipeline (Section 5) could be paired with these reagents in a completed partitioning assay, this remains the group's immediate next step.

Conclusions and Future Directions

Taken together, this work provides converging, if not yet complete, evidence that the Nup62 FG-repeat IDR of the NPC central channel is under more specific, sequence-level and biochemical constraints than its classification as 'intrinsically disordered' would suggest. The FG motifs themselves, not the

surrounding disordered sequence generally, are preferentially conserved across vertebrates; the domain undergoes salt-modulated condensation, first identified as an unexpected purification anomaly and then confirmed quantitatively with a fluorescence-independent assay developed specifically to overcome the protein's lack of a chromophore; and the resulting assemblies appear more aggregate-like than liquid-like under the single set of conditions tested so far a distinction the BSA/PEG positive control shows is a genuine biological result rather than a methods artifact. Establishing a reversed-phase HPLC/lyophilization route to concentrate the domain and purifying NTF2 and mEGFP together provides the reagents and protocols needed to directly test the phase-separation selectivity hypothesis this thesis set out to address.

Concrete next steps:

- Repeat and extend the in vitro condensate-formation experiment (Section 6) across a range of protein concentration, temperature, and crowding agents, to determine whether liquid (fusion-competent, FRAP-recoverable) droplets can be obtained, or whether aggregation is the domain's intrinsic behavior in isolation.
- Test the functional significance of the conserved FG positions directly, by mutating conserved versus variable repeats and re-running the phase-separation assay.
- Complete the selective-partitioning assay with the purified NTF2 and mEGFP reagents, to test whether Nup62 condensates (liquid or aggregate) retain NPC-like cargo selectivity — the central open question motivating this project.

Techniques and Skills Developed

- Computational identification of intrinsically disordered regions and FG-repeat motifs; multiple sequence alignment and per-residue conservation scoring across vertebrate species
- Custom, code-based sequence analysis (amino acid adjacency / flanking-residue frequency analysis)
- Recombinant expression and purification of an aggregation-prone, chromophore-free intrinsically disordered protein (Ni-NTA and size-exclusion chromatography)
- Reversed-phase HPLC purification and lyophilization (SpeedVac) to concentrate a hydrophobic IDR to millimolar concentrations
- SDS-PAGE and densitometric quantification (ImageJ) as a substitute for UV-based protein quantification
- Design and execution of a custom centrifugation-based phase-separation assay
- Brightfield/DIC and fluorescence microscopy of biomolecular condensates, including fluorophore labeling and negative-control design
- Critical evaluation of experimental design choices in the LLPS field (e.g., hydrogel vs. condensate assay formats) grounded in primary literature